

Data Visualization Types using Databricks SQL

Dataset Used: samples.tpch.orders

Query: select * from samples.tpch.orders

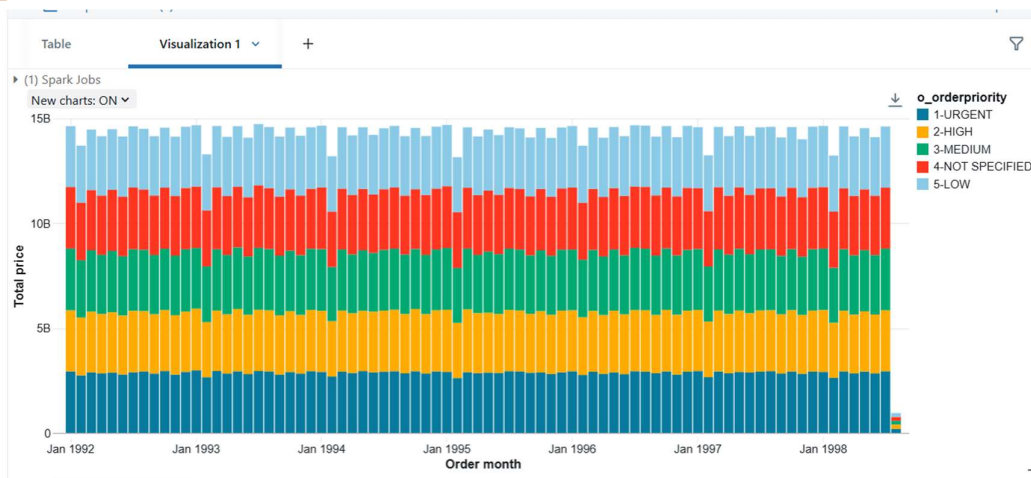
1. Bar Chart

Purpose: Used to compare values across different categories.

Configuration values: For this bar chart visualization, the following values were set:

- X column:
 - Dataset column: o_orderdate
 - Date level: Months
- Y columns:
 - Dataset column: o_totalprice
 - Aggregation type: Sum
- Group by (dataset column): o_orderpriority
- Stacking: Stack
- X axis name (override default value): Order month
- Y axis name (override default value): Total price

Screenshot:



2. Line Chart

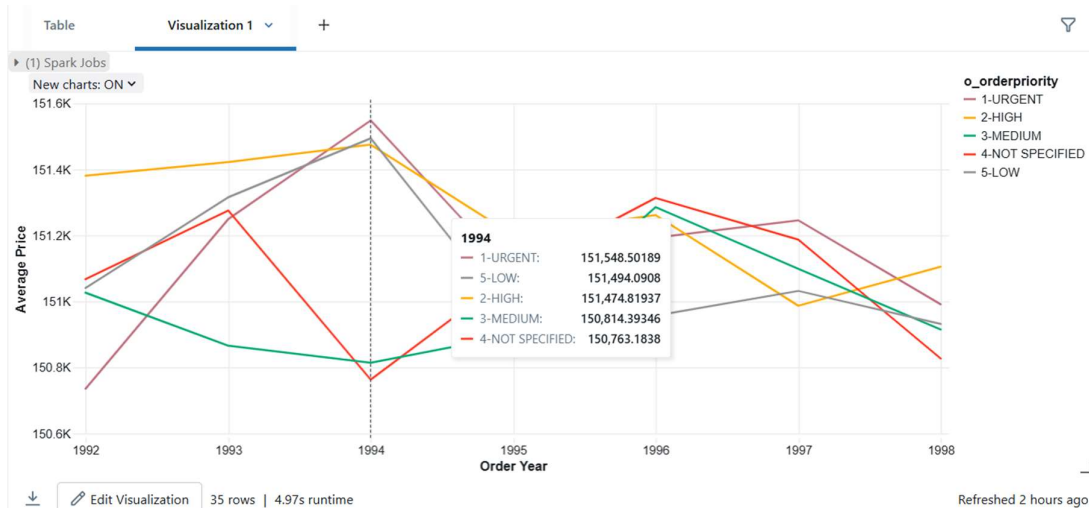
Purpose: Shows trends over time.

Configuration values: For this line chart visualization, the following values were set:

- X column:
 - Dataset column: o_orderdate
 - Date level: Years
- Y columns:
 - Dataset column: o_totalprice
 - Aggregation type: Average
- Group by (dataset column): o_orderpriority
- X axis name (override default value): Order year
- Y axis name (override default value): Average price



Screenshot:



3. Area Chart

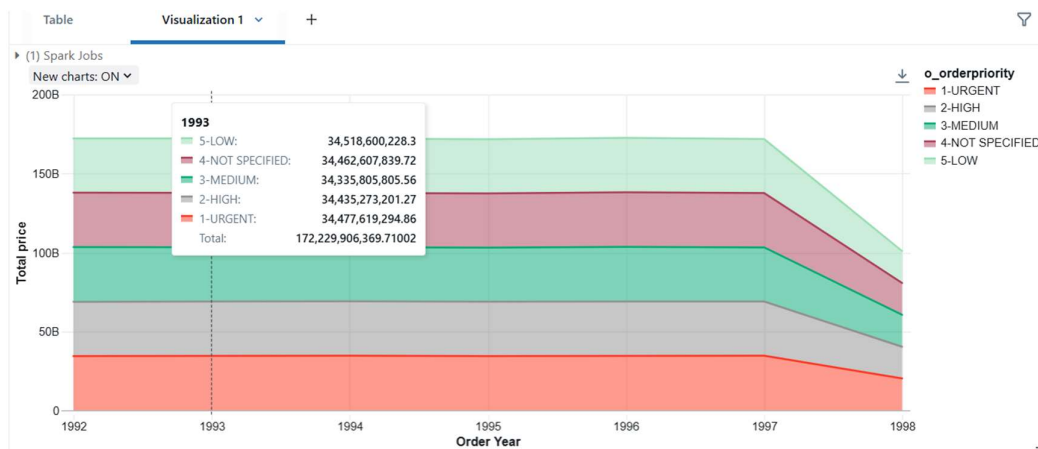
Purpose: Like a line chart but with the area below the line filled.

Configuration values: For this area chart visualization, the following values were set:

- X column:
 - Dataset column: o_orderdate
 - Date level: Years
- Y columns:
 - Dataset column: o_totalprice
 - Aggregation type: Sum
- Group by (dataset column): o_orderpriority
- Stacking: Stack
- X axis name (override default value): Order year
- Y axis name (override default value): Total price



Screenshot:



4. 🥞 Pie Chart

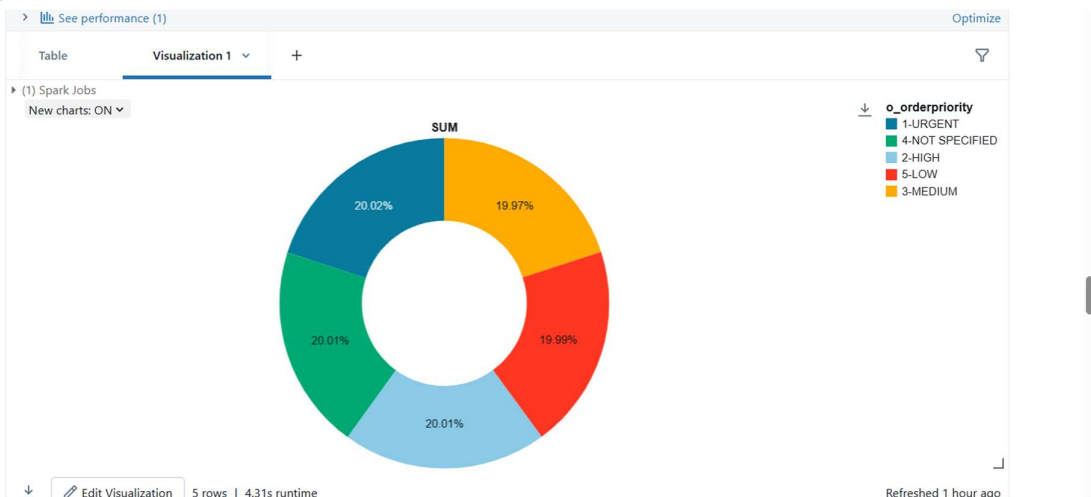
Purpose: Shows proportions/percentages in a whole.

Configuration values: For this pie chart visualization, the following values were set:

- X column (dataset column): o_orderpriority
- Y columns:
 - Dataset column: o_totalprice
 - Aggregation type: Sum
- Label (override default value): Total price



Screenshot:



5. 📊 Histogram

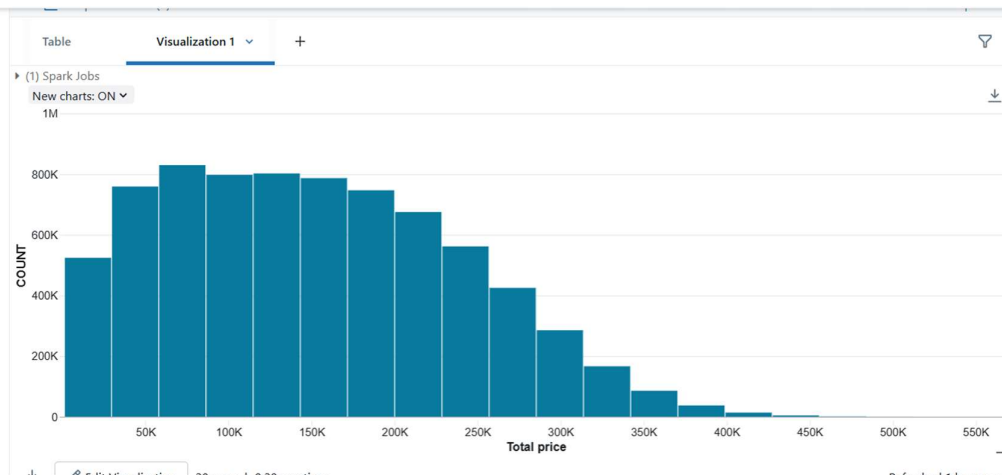
Purpose: Shows distribution of a numeric value.

Configuration values: For this histogram chart visualization, the following values were set:

- X column (dataset column): o_totalprice
- Number of bins: 20
- X axis name (override default value): Total price



Screenshot:



6. 📈 Heat Map

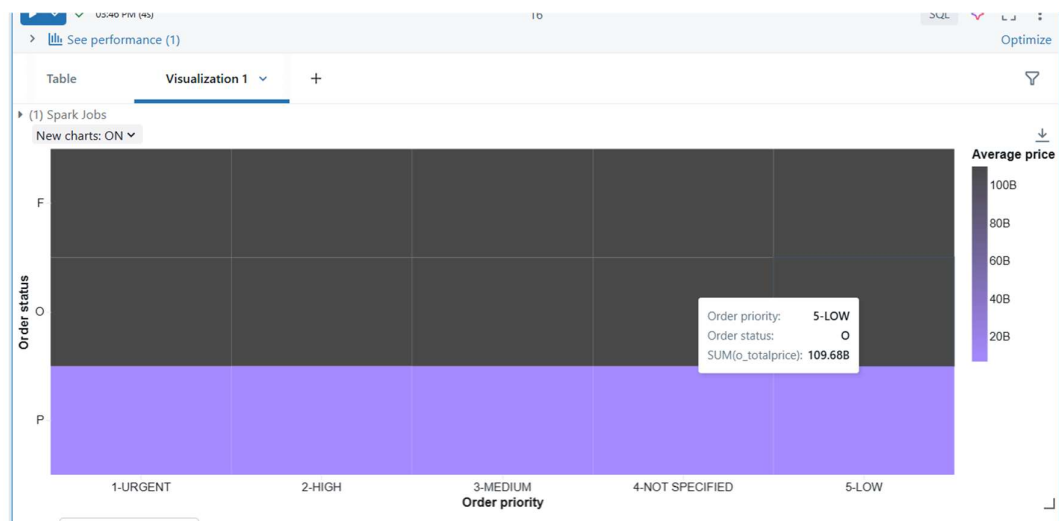
Purpose: Shows data density or correlation using colors.

Configuration values: For this heatmap chart visualization, the following values were set:

- X column (dataset column): o_orderpriority
- Y columns (dataset column): o_orderstatus
- Color column:
 - Dataset column: o_totalprice
 - Aggregation type: Average
- X axis name (override default value): Order priority
- Y axis name(override default value): Order status
- Color name (override default value): Average price
- Color scheme (override default value): YlGnBu



Screenshot:



7. Scatter Chart

Purpose: Shows relationship between two numeric variables.

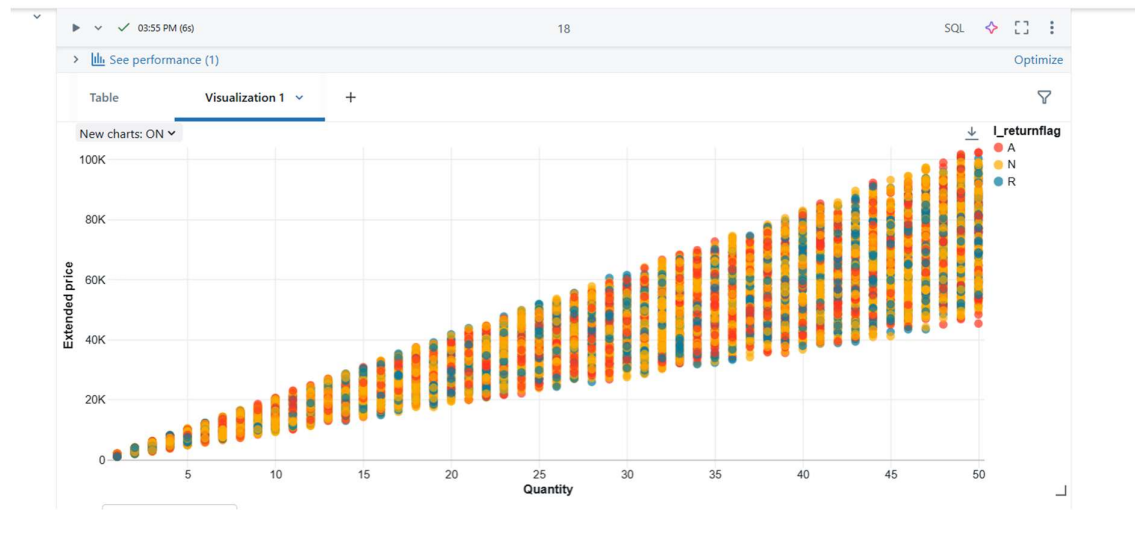
Configuration values: For this scatter chart visualization, the following values were set:

- X column (dataset column): l_quantity

- Y column (dataset column): l_extendedprice
- Group by (dataset column): l_returnflag
- X axis name (override default value): Quantity
- Y axis name (override default value): Extended price



Screenshot:



8. ● Bubble Chart

Purpose: Similar to scatter plot but includes a third variable as bubble size.

Configuration values: For this bubble chart visualization, the following values were set:

- X (dataset column): l_quantity
- Y columns (dataset column): l_extendedprice
- Group by (dataset column): l_returnflag
- Bubble size column (dataset column): l_tax
- Bubble size coefficient: 20
- Bubble size proportional to: Area
- X axis name (override default value): Quantity
- Y axis name (override default value): Extended price



Screenshot:

